Effect Of Load On Bar, Body And System Power Output In The Power Clean

Tracie Haines, Jeffrey M. McBride, Jared Skinner, Mark Woodall, Tony R. Larkin, Tyler J. Kirby, Andrea M. Dayne

Various methods of calculation can affect power output values in the power clean. This may be due to analysis of the bar, body and system using different kinetic and kinematic variables. PURPOSE: The purpose of this study was to utilize the combination of a force plate and videography to determine power output of the bar, the body, and the system independent of one another. METHODS: Seven college-aged males (height = 175.29±5.47 cm, weight = 80.84±7.18 kg, age = 24.7±2.06 yr, 1RM = 97.14±6.36 kg) with at least one year experience in the power clean performed two sets of one repetition each at 30, 40, 50, 60, 70, 80, and 90% of their 1 repetition maximum (1RM) in a randomized order. Force, power, and velocity were obtained for the bar, body and system independently. RESULTS: Peak power (PP) of the bar was found to be at 90% of 1RM (2308±229.1 W), PP of the body at 90% of 1RM (1077±538.7 W) and PP of the system at 80% of 1RM (1768±470.7 W). Significant differences ($p \leq 0.05$) in PP were found between the bar and body at 40%, 50%, 60%, and 70% of 1RM, while 60% of 1RM also showed significant differences between the system and body. Significant differences between the bar, body and system occurred at 80% and 90% of 1RM for PP. Peak velocity (PV) occurred for the bar, body and system at 30% of 1RM (Bar PV 30% = 2.38±0.15 m/s), 30% of 1RM (Body PV 30% = 0.79±0.21 m/s) and 60% of 1RM (System PV 60% = 0.90±0.16 m/s), respectively. Every load displayed significant differences in PV between the bar and body and between the bar and system, except for 60% of 1RM, where a significant difference occurred between the bar, body, and system. Peak force (PF) was highest at 90% of 1RM for the bar (1291±72.9 N), while both the body and system PF occurred at 80% of 1RM (Body PF 80% = 1505±270.1 N, System PF 80% = 2797±618.2 N). Significant differences in peak force were found between the bar, body and system at every load, except 90% of 1RM where significant differences were between the bar and body and between the bar and system. CONCLUSION: In conclusion, bar, body, and system values for PP, PV, and PF are influenced differently across the loading spectrum. PRACTICAL APPLICATION: Proper training loads for the power clean maybe influenced by whether PP of the bar, body or system is desired.